Week 7 In-Class Worksheet
Nonparametric Tests

Name: ______________________________ **Date:** ______________

Course: H524 Introduction to Biostatistics
Topic: Sign Test, Wilcoxon Tests, Kruskal-Wallis, Normality Assessment

Instructions

Work with a partner to solve the following problems. Show all your work. Use R to perform tests and verify your calculations.

1 Problem 1: Sign Test

A physical therapist wants to test if a new exercise program reduces recovery time. The therapist believes the median recovery time should be less than 14 days. Recovery times (days) for 10 patients were:

12, 15, 10, 13, 16, 11, 14, 9, 12, 15

Part A: State the null and alternative hypotheses.

Part B: Calculate the number of positive signs (B^+), negative signs (B^-), and the test statistic B.

Part C: Use R to calculate the p-value for this one-sided test.
R code:

P-value: ____________

Part D: What is your conclusion at $\alpha = 0.05$?

2 Problem 2: Wilcoxon Signed-Rank Test

Using the same recovery time data from Problem 1, perform a Wilcoxon signed-rank test to determine if the median recovery time differs from 14 days.

Part A: Calculate the differences from 14 days and rank the absolute differences (excluding zeros).

Part B: Calculate the sum of positive ranks (W^+) and negative ranks (W^-).

Part C: Use R to perform the test and find the p-value.
R code:

P-value: ________________

Part D: Compare the p-value to the sign test. Which test is more powerful here?

3 Problem 3: Paired Data

A study measured systolic blood pressure (mmHg) before and after a stress reduction program for 8 participants:

Participant	1	2	3	4	5	6	7	8
Before	142	138	155	148	160	145	152	149
After	138	140	148	145	155	142	150	145

Part A: Calculate the differences (Before - After).

Part B: Perform a sign test. What is the p-value?
R code:

P-value: ______________

Part C: Perform a Wilcoxon signed-rank test. What is the p-value?
R code:

P-value: ______________

Part D: Which test would you recommend for these data? Why?

4 Problem 4: Two-Sample Wilcoxon Rank-Sum Test

A researcher compares pain relief scores (0-10 scale) for two different medications:

Medication A: 6, 7, 5, 8, 6, 7, 9
Medication B: 4, 5, 6, 5, 7, 4, 6

Part A: Use R to perform the Wilcoxon rank-sum test (Mann-Whitney U test).
R code:

W statistic: ______________ **P-value:** ______________

Part B: What do you conclude at $\alpha = 0.05$?

Part C: Why might you choose this test instead of a two-sample t-test?

5 Problem 5: Kruskal-Wallis Test

A clinical trial compares three different dosages of a sleep medication by measuring hours of sleep:

Low dose: 5.2, 6.1, 5.8, 6.3, 5.5
Medium dose: 6.8, 7.2, 7.5, 6.9, 7.1
High dose: 7.8, 8.2, 8.5, 7.9, 8.1

Part A: Use R to perform the Kruskal-Wallis test.
R code:

H statistic: ______________ **P-value:** ______________

Part B: What do you conclude at $\alpha = 0.05$?

Part C: If significant, what follow-up test should you perform?

6 Problem 6: Checking Normality Assumptions

For the high dose group from Problem 5:

Part A: Use the Shapiro-Wilk test to check normality.
R code:

W statistic: ______________ **P-value:** ______________

Part B: Can you assume normality? Explain.

Part C: Create a Q-Q plot in R. What does it show?
R code:

Part D: Based on this, was the Kruskal-Wallis test appropriate for Problem 5?

7 Problem 7: Choosing the Right Test

For each scenario, state which test you would use and briefly explain why:

Part A: Comparing median survival times (highly skewed) between two treatment groups (n=15 each).

Part B: Testing if the median number of hospital readmissions differs from 3 in a sample of 20 patients.

Part C: Comparing average weight loss across four different diet programs (n=10 per group, data appears normal).

Part D: Comparing pre- and post-intervention depression scores for 12 patients (ordinal scale 1-10).

R Code Reminders:

- **Sign Test:** `binom.test(n_positive, n_total, p=0.5)`
- **Wilcoxon Signed-Rank:** `wilcox.test(data, mu=value)`
- **Wilcoxon Rank-Sum:** `wilcox.test(group1, group2)`
- **Kruskal-Wallis:** `kruskal.test(outcome ~ group)`
- **Shapiro-Wilk:** `shapiro.test(data)`